

TECHNICAL DESIGN DOCUMENT

Diagnostic-Grade ECG Signal Processing & Acquisition

AI Circuit Breaker Technical Implementation Framework

An Observability-First Engineering Framework for IoT-Enabled Cardiovascular Diagnostics

Integrating Diagnostic Signal Fidelity with AI Trust Metrology, Semantic Governance, and Deterministic Safety Controls for Trustworthy Autonomous Cardiac Analysis

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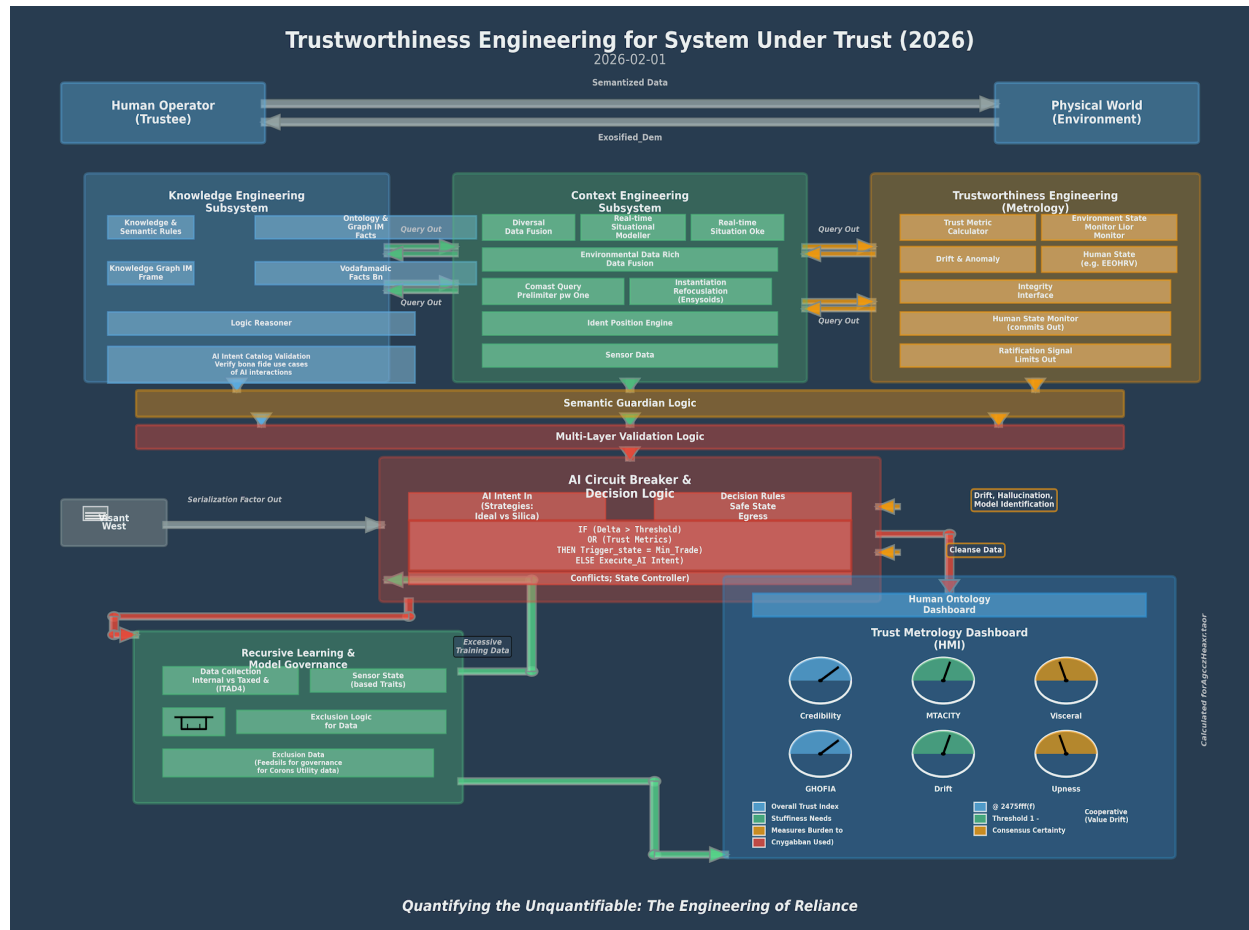


Figure 0.1 — Trustworthiness Engineering for System Under Trust: Complete Architecture (2026)

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1. Document Purpose and Scope

This Technical Design Document (TDD) defines the complete engineering specification for a Diagnostic-Grade Electrocardiogram (ECG) Signal Processing and Acquisition system integrated with an AI Circuit Breaker and Trust Engine. It formally unifies two architectural paradigms: the precision signal fidelity requirements mandated by the American Heart Association (AHA) and ANSI/AAMI standards, and the Observability-First trustworthiness engineering framework that treats AI reliability as a measurable, auditable, and engineered property.

The fundamental premise of this design is that diagnostic ECG interpretation powered by AI cannot be treated as a black box. Each probabilistic assertion generated by the AI classification layer must be continuously vetted against the physical ground truth of the cardiac signal. This document establishes the technical mechanisms by which that vetting is performed, how trust is quantified across the full device lifecycle, and how the system defaults to a safe state when the gap between AI intent and physical reality exceeds a clinically defined tolerance.

1.1 Scope of Application

- **Design Time:** Manufacturing, algorithm training, and semantic governance during device development and regulatory validation.
- **Run Time – Edge:** Real-time patient monitoring via IoT-enabled portable and wearable ECG devices.
- **Run Time – Clinical:** Cloud-based supervisory agents integrated with enterprise EHR dashboards and cardiology workflows.
- **Lifecycle Governance:** Continuous learning pipelines with recursive sanitization to prevent model collapse.

1.2 Governing Standards and References

Standard / Reference	Application in This Design
ANSI/AAMI EC11 / IEC 60601-2-25	ECG hardware safety, performance bandwidth, and safety classification
AHA/ACC ECG Standards	Sampling rates, filtering, measurement accuracy, compression, and display
MIT-BIH / AAMI Arrhythmia DB	Beat classification taxonomy (N, S, V, F, Q) and model validation benchmarks
W3C RDF/OWL	Semantic ontology for ECG impossible states and Knowledge Fabric guardrails
ISO/IEC/IEEE 24765	Systems engineering lifecycle governance and change management
NIST AI RMF (2026)	AI risk management framework; trustworthiness measurement and auditing
SHACL / KGCL / ORSD	Knowledge graph validation, change management, and full audit trails

2. Strategic Foundations: The Observability-First Paradigm

Modern cardiovascular diagnostics sit at the intersection of two irreversible technological transitions: the permanent shift from analog galvanometric acquisition to high-frequency digital systems, and the emergence of AI-driven clinical interpretation. Each transition offers transformative diagnostic capability and introduces a distinct category of failure risk. This design addresses both simultaneously through an Observability-First engineering philosophy that subordinates AI autonomy to measurable, verifiable trustworthiness.

The traditional Performance-First mindset—prioritizing model speed and accuracy at the expense of transparency—is clinically unacceptable in cardiac diagnostics. A high-confidence misclassification of atrial fibrillation, a missed STEMI, or a false-positive pacemaker capture can directly result in patient harm. The Observability-First framework reorients the design around a single foundational principle: every AI assertion about a patient’s cardiac rhythm must be traceable to a deterministic, physically grounded source of truth.

2.1 The Metrology of Trust

The Metrology of Trust treats trustworthiness as a measurable physical property of the system, engineered with the same rigor applied to signal amplitude accuracy or filter bandwidth. Within this framework, the fundamental question shifts from “Is the AI smart?” to “Is the AI trustworthy?” across four operational dimensions:

Dimension	Definition	ECG Implementation
Measurability	Trust quantified via high-precision metrics: drift, semantic anomalies, situational awareness.	Semantic Anomaly Score for rhythm classifications; Contextual Relevancy Index for electrode contact quality.
Verification	AI intent is continuously audited against W3C RDF/OWL ontologies defining physiological possibility.	Formalized Impossible States (HR > 300 bpm intrinsically; PR < 60 ms) constrain the AI inference layer.
Calibration	Real-time calibration monitors the human operator’s physiological state alongside AI health indicators.	Human-AI Calibration Coefficient adjusts CB thresholds based on cardiologist cognitive load (HRV/EEG proxy).
Observability	Decisions are traced to deterministic sources of truth; human comprehension is anchored as complexity scales.	All CB events logged with full trace: signal epoch, AI assertion, delta value, trust scores, and safe state action.

2.2 The Ground Truth Circuit Breaker as Clinical Safety Gate

The foundational safety element of this architecture is the Ground Truth Circuit Breaker. In the ECG domain, ground truth is defined by the electrochemical signal captured at the electrode-skin interface—a deterministic measurement of myocardial depolarization and repolarization governed by immutable laws of cardiac electrophysiology. This physical signal is

the absolute authority over any AI-generated interpretation. The circuit breaker serves as the non-negotiable enforcement mechanism preventing any AI inference from overriding physical reality without triggering a clinically appropriate safe state.

Design Principle — The Deterministic Veto

The ECG electrode signal constitutes ground truth. No AI classification assertion may be acted upon clinically if its Intent Delta exceeds the Sensor Ground Truth Tolerance. The circuit breaker is not advisory—it is a deterministic veto with absolute authority over all autonomous AI actions.

3. System Architecture: Trustworthiness Engineering Model

The integrated system architecture is organized into six interconnected subsystems as depicted in the Trustworthiness Engineering model below. Each subsystem corresponds to a distinct functional domain with clearly defined data flows, control interfaces, and trust measurement points. The AI Circuit Breaker and Decision Logic subsystem sits at the center of the architecture, receiving inputs from all other subsystems and exercising deterministic veto authority over the AI output pipeline.

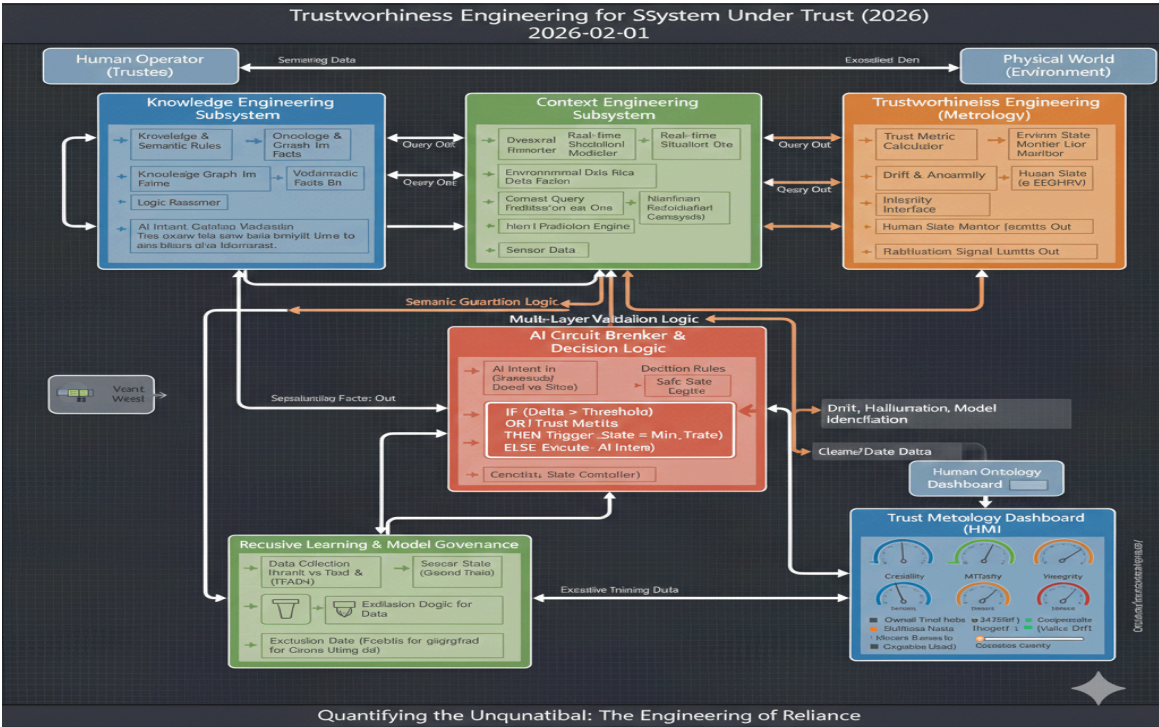


Figure 3.1 — System Architecture Overview: Trustworthiness Engineering for ECG System Under Trust

3.1 Six-Subsystem Overview

Subsystem	Primary Function	ECG Domain Role
Knowledge Engineering	Semantic modeling, ontology management, knowledge graph storage, and logic reasoning.	Encodes clinical cardiology ontologies; defines physiologically impossible states as machine-readable guardrails.
Context Engineering	Environmental context fusion, patient data retrieval, and situational awareness integration.	Integrates patient history, electrode placement flags, medication context, and interference data.
Trustworthiness Engineering (Metrology)	Trust metric computation, drift detection, and conformance assessment against standards.	Computes Semantic Anomaly Score, Contextual Relevancy Index, MTBH, and Value-Drift Coefficient.

AI Circuit Breaker & Decision Logic	Trip logic evaluation, deterministic veto execution, and safe state activation.	Core enforcement layer; evaluates every AI ECG output against physical sensor ground truth in real time.
Recursive Learning & Model Governance	Continuous model training, negative exclusion filtering, and vector DB knowledge updates.	Purges circuit-breaker-flagged epochs; governs classifier updates; prevents model collapse.
Trust Metrology Dashboard	Real-time visualization of Credibility, Validity, Viability, and Overall Trust Index.	Clinical console for cardiologist oversight, trustee escalation, and system health monitoring.

3.2 Four-Layer Architectural Stack

The signal-to-decision pipeline traverses four discrete processing layers, each with its own source of truth and its own failure mode. The AI Circuit Breaker operates as a cross-cutting enforcement mechanism across all four layers simultaneously.

Layer	Function	Source of Truth	Primary Failure Mode
Data Layer	Raw ECG acquisition via electrodes, ADC, and high-frequency oversampling front-end.	Physical electrochemical signal at the skin-electrode interface.	Sensor noise, motion artifacts, electrode displacement, and powerline interference.
Algorithmic Layer	Digital filtering, template construction, global interval measurement, and feature extraction.	Deterministic DSP algorithms with fixed mathematical operations.	Improper filter selection, temporal misalignment, and compression artifacts.
LLM / AI Layer	Rhythm classification, arrhythmia detection, ST interpretation, and clinical statement generation.	Probabilistic inference over learned feature distributions.	Super-Huberous assertions: high-confidence misclassifications lacking physical grounding.
Circuit Breaker Layer	Continuous delta evaluation, ontology verification, and safe state enforcement.	Deterministic veto rules and W3C RDF/OWL cardiac domain ontologies.	Misconfigured tolerance thresholds; incomplete ontology coverage for novel patterns.

4. Signal Acquisition Layer: Digital Front-End Standards

High-fidelity signal acquisition is the non-negotiable bedrock of the entire architecture. The Data Layer constitutes the physical ground truth upon which the AI Circuit Breaker exercises its deterministic veto authority. If the signal entering the pipeline is corrupted by sensor noise, aliasing, or electrode artifacts, the circuit breaker cannot validate against a reliable baseline. Acquisition standards are therefore treated as the zero-tolerance foundation of the entire trustworthiness framework.

4.1 Oversampling and Processing Rate Requirements

The industry has executed a permanent strategic shift from analog galvanometric systems to high-frequency digital acquisition architectures. This transition is a prerequisite for the AI processing pipeline, enabling both anti-aliasing and the data density required for precision digital filtering.

- **Front-End Acquisition Rate:** 1,000 to 15,000+ samples per second (Hz). Oversampling prevents aliasing of narrow, high-frequency components such as pacemaker spikes at < 0.5 ms duration, and provides the data density required for downstream precision filtering.
- **Processing / Output Sampling Rate (Minimum):** 500 Hz. The Nyquist theorem and AHA recommendations mandate a minimum of 500 Hz to maintain amplitude measurement errors below 1% for all clinically relevant frequency content.
- **Rule of Ten (SOP-01 Prerequisite):** Sensor precision must be at least one order of magnitude higher than the operational tolerance threshold. For a 10 μ V AHA amplitude fidelity mandate, the ADC must operate with a signal resolution of ≤ 1 μ V.

4.2 Pacemaker Pulse Detection Standards

Pacemaker pulse detection represents one of the most demanding acquisition requirements in the system. Modern bipolar pacemakers deliver stimuli with durations as short as 0.1–0.5 ms, which standard processing rates cannot reliably capture. Failure to detect these pulses, or misrepresentation of their amplitude, creates a primary circuit breaker trip condition by generating a morphological mismatch between AI classification and the physical signal.

Parameter	Acquisition Requirement	Display & CB Constraint
Stimulus Duration	System must reliably detect pulses < 0.5 ms in duration.	Artificial amplitude enhancement of the spike is strictly forbidden. CB flags any amplified spike as a Semantic Anomaly.
Sampling Minimum	Front-end oversampling at a minimum of 10,000 samples/sec is mandatory for reliable capture.	SOP-01 Handshake verifies acquisition rate before autonomous AI operation is permitted.
Pulse Preservation	Identification of atrial, ventricular, and biventricular spikes is	Digital markers shall distinguish pacing signals from intrinsic QRS

	maintained through storage and retrieval.	complexes without altering the raw trace.
CB Integration	If AI assigns a QRS label to a pacemaker artifact epoch, the Semantic Anomaly Score is elevated, and CB evaluation is triggered.	Trip event logged; data excluded from training queue (SOP-03). Physician overread the mandatory.

4.3 Circuit Breaker Integration: SOP-01 Semantic Handshake

The circuit breaker monitors the acquisition layer in two ways. First, it executes a pre-operation Semantic Handshake at initialization to verify that sensor precision meets the Rule of Ten threshold. Second, it continuously evaluates the Contextual Relevancy Index to detect electrode contact loss, impedance drift, and environmental interference.

```
SOP-01: SEMANTIC HANDSHAKE (Acquisition Pre-Flight)
Step 1: Ping Knowledge Graph → verify ECG ontology is current and consistent
Step 2: IF (ADC_Resolution > Operational_Tolerance / 10)
    THEN PROHIBIT autonomous operation
    ALERT: "Sensor precision insufficient – manual cardiologist review required"
Step 3: Impedance_Sweep(all_electrodes)
    IF any(electrode_impedance > 5 kΩ) OR lead_off_flag THEN PROHIBIT
Step 4: Verify diagnostic filters (LPF 150/250 Hz; HPF 0.05/0.67 Hz active)
Step 5: Initialize trust gauges to baseline; log initialization state with timestamp
```

5. Digital Filtering Protocols: Preserving Morphological Integrity

The strategic selection of bandwidth is paramount. Improper filtering introduces diagnostic artifacts that directly cause misinterpretation of ischemia, infarction, and conduction abnormalities. The filtering stage bridges raw acquisition data and the AI classification layer; any morphological distortion introduced here propagates upward as a processing artifact that the LLM layer may interpret as a pathological signal—a classic Super-Huberous failure mode. The circuit breaker must therefore validate the active filter state as a primary component of the Contextual Relevancy Index.

5.1 Low-Frequency (High-Pass) Filter Specifications

The high-pass filter suppresses baseline wander caused by respiratory motion and perspiration. Its cutoff must be precisely controlled to avoid distorting the ST-segment, the primary electrocardiographic indicator of myocardial injury.

- **Standard Diagnostic Cutoff:** 0.05 Hz. Mandated for all routine resting 12-lead ECG systems.
- **Relaxed Cutoff for Zero-Phase Systems:** 0.67 Hz, permissible only when linear digital filters with zero phase distortion are implemented via bidirectional filtering or flat step response designs.
- **Engineering Rationale:** Zero-phase filters eliminate phase nonlinearities that cause artifactual ST-segment deviation. A system using a 0.67 Hz cutoff without zero-phase implementation will systematically produce false ST changes, leading to high-confidence AI misclassification of ischemia.

Circuit Breaker Trigger — High-Pass Filter Violation

If the active high-pass cutoff exceeds 0.67 Hz during a diagnostic recording without verified zero-phase implementation, the Semantic Anomaly Score is elevated and a mandatory user alert is generated. Any AI-generated ST-segment interpretation produced during this state is excluded from clinical action and training data.

5.2 High-Frequency (Low-Pass) Filter Specifications

The low-pass filter determines the system’s ability to capture the high-frequency components of the QRS complex, essential for the detection of Q-waves (myocardial infarction) and QRS notching (bundle branch block). Standards are population-stratified due to the intrinsically higher frequency content of pediatric cardiac signals.

Population	Mandated LPF Cutoff	Clinical Rationale
Adults and Adolescents	150 Hz	Maintains amplitude errors below 1%. Accurate detection of Q-waves and notched QRS for MI and LBBB/RBBB diagnosis.

Infants and Children	250 Hz	Required due to significantly higher QRS frequency content in pediatric signals. 150 Hz smooths critical features in children.
Monitoring Mode (PROHIBITED for diagnostics)	40 Hz	CLINICAL INVALIDATION: Smooths diagnostic notches; invalidates amplitude measurements. Must never be active during diagnostic recordings.

5.3 Filter State in the Circuit Breaker

```
CONTEXTUAL RELEVANCY INDEX - Filter State Evaluation:
IF (Active_LPF_Cutoff < 150 Hz AND Population == ADULT)
THEN CRI -= 0.30  TRIGGER alert: "Monitoring filter active - diagnostic data invalid"
IF (Active_HPF_Cutoff > 0.67 Hz AND ZeroPhase_Verified == FALSE)
THEN CRI -= 0.50
TRIGGER circuit breaker evaluation for ST-segment assertions
IF (CRI < CRI_MINIMUM_THRESHOLD)
THEN ACTIVATE CIRCUIT BREAKER (filter-induced context failure)
EXECUTE Safe_State_Level2()
```

6. Waveform Synthesis and Global Measurement Logic

The evolution from sequential single-channel recording to simultaneous 12-lead acquisition fundamentally changes the nature of ECG measurement. In single-lead systems, interval measurements are limited by the spatial perspective of one lead vector. In global measurement systems, every waveform is measured across all temporally aligned leads simultaneously, providing complete spatial coverage of the cardiac electrical vector. This transition is critical both for diagnostic accuracy and for the circuit breaker’s ability to assess measurement plausibility against ontologically defined interval boundaries.

6.1 Time Coherence: The Critical Differentiator

Time Coherence is the precise temporal alignment of all 12 leads at every sampling point—the prerequisite for all valid global measurements and for the circuit breaker’s cross-lead consistency checks. Systems must achieve temporal alignment accurate to within ± 10 ms across all leads. Misalignment above this threshold introduces artificial interval variability that the AI layer may misinterpret as conduction abnormalities.

6.2 Representative Complex (Template) Construction

- Beat Alignment:** Cross-correlation algorithms must be used to match identical points in the cardiac cycle across all leads, ensuring the template reflects a true spatial snapshot of the cardiac vector.
- Selective Averaging:** The logic must differentiate between dominant beats (sinus rhythm) and nondominant beats (ectopics, PVCs). Ectopic beats must be excluded from the dominant template to prevent artificial QRS widening.
- Mean vs. Median Templates:** Median templates are preferred for their ability to exclude transient impulse noise without requiring explicit beat classification.

6.3 Global Measurement Logic: Earliest Onset / Latest Offset Rule

Modern systems must utilize global measurement logic rather than single-lead analysis for all interval and duration measurements. The global approach recognizes that a single lead may be perpendicular to the heart vector at the onset or offset of a waveform, causing it to appear isoelectric. At the same time, the signal is clearly present in other leads.

Measurement Domain	Global Rule	Circuit Breaker Implication
All Waveform Durations	System searches all aligned leads to identify the absolute earliest start and latest end of every waveform (P-wave, QRS, T-wave, QT interval).	AI AV block or wide QRS classifications validated against global (not single-lead) intervals. Divergence triggers Semantic Anomaly elevation.
PR Interval / AV Block	Global PR intervals are systematically longer than single-lead measurements. Normal	AI assertions of first-degree AV block verified against global PR before

	ranges must be recalibrated for global measurement standards.	clinical action. Systematic discrepancy flagged by the CB.
Q-Wave Duration / MI	The Earliest Onset rule is critical. Underestimating Q-wave duration by failing to identify its earliest onset in any lead produces a false-negative MI diagnosis.	An AI “normal” classification, contradicting a globally significant Q-wave, triggers mandatory CB evaluation and physician overread.

7. Data Compression and Storage Fidelity

As ECG data migrates from bedside acquisition to the Electronic Health Record and cloud-based analysis platforms, a strategic balance between storage efficiency and diagnostic accuracy must be maintained. Lossy compression poses a direct risk to high-frequency signal components—the components most diagnostically critical and most vulnerable to degradation. The AI circuit breaker must validate data integrity at both the point of compression and the point of retrieval before authorizing AI classification of any archived record.

7.1 ECG Data Compression Methods and Trade-offs

Technique	Ratio	Fidelity Impact	CB / Trust Implications
FFT / Discrete Cosine Transform	8:1 to 10:1	Low error (0.5%–2%); may smooth high-frequency QRS notches.	Acceptable within AHA mandate if error < 10 μ V. CB performs an integrity check before AI classification.
Wavelet Transforms	Up to 20:1	Higher potential error (up to 4%); requires high-fidelity implementation.	Error > 10 μ V triggers CB block and lossless re-acquisition requirement.
Bimodal Decimation	Segment-variable	Maintains QRS at 500 sps; decimates P-waves and T-waves.	Preferred for pacemaker-critical systems. P-wave AI analysis requires separate high-fidelity sampling track.
Lossless (Future State)	2:1 to 4:1	Zero amplitude error. Original and retrieved tracings are mathematically identical.	Only method eliminating compression-induced CB trips. Mandated where bandwidth and storage permit.

7.2 The AHA Fidelity Mandate and CB Integrity Enforcement

AHA Fidelity Mandate (Non-Negotiable)

The maximum allowable error for any retrieved compressed ECG data is 10 μ V from the original signal. This mandate applies to all waveform segments without exception. High-frequency signals—specifically pacemaker spikes and rapid R-wave upstrokes—are the components most susceptible to compression-induced degradation.

```
COMPRESSION INTEGRITY CHECK (Pre-AI Classification):
FOR each retrieved ECG archive record:  delta_max = MAX(|Original_Sample[i]
- Retrieved_Sample[i]| for all i)  IF (delta_max > 10  $\mu$ V)      THEN BLOCK AI
classification                    FLAG record: compression_integrity_violation
LOG: epoch, delta_error, compression_method, record_ID          REQUIRE:
lossless re-acquisition OR manual cardiologist overread
```

8. Clinical Hardware Implementation and Display Standards

Digital signal processing precision cannot compensate for failures at the physical-to-digital interface. The electrode-to-skin interface is the single most common source of clinical diagnostic error in ECG acquisition and represents the ground floor of the circuit breaker’s physical ground truth. The hardware implementation standards below define the physical conditions that must be verified before the AI interpretation pipeline is authorized to operate.

8.1 Electrode Placement Standards

Systematic electrode misplacement introduces persistent, reproducible morphological distortions that the AI layer will interpret as pathology with high confidence—the definition of a Super-Huberous failure. These are particularly dangerous because their consistency prevents the circuit breaker’s pattern-of-error detection from flagging them as transient artifacts.

- **Precordial Leads V1–V6 (Horizontal Placement Rule):** V1 and V2 must be placed in the fourth intercostal space at the right and left sternal borders. V4 must be at the fifth intercostal space at the midclavicular line. V5 and V6 must be placed along the strict horizontal plane passing through V4—not along the intercostal space, which curves inferiorly. Following the intercostal space for V4–V6 placement is strictly forbidden, as it introduces systematic inferior displacement that mimics inferior ST changes.
- **Mason-Likar (Torso) Placement:** Torso-placed limb electrodes are not interchangeable with standard resting 12-lead ECGs. Torso placement distorts the Wilson Central Terminal and significantly alters the QRS axis and R-wave progression. Any ECG acquired with torso placement must be metadata-tagged so the AI classification engine loads modified torso-placement reference ranges before classifying.

CB Placement Validation at SOP-01

At initialization, the system performs an automated impedance sweep across all 10 electrodes. If any electrode impedance exceeds 5 kΩ, or if lead-off detection flags are active, autonomous operation is prohibited. The CRI is reduced by 0.40 per non-compliant electrode.

8.2 Display Speed Standards

Display Speed	1 mm Small Square	5 mm Large Square	Clinical Utility
25 mm/sec (Standard)	0.04 sec (40 ms)	0.20 sec (200 ms)	Standard resting diagnosis; global interval measurement.
50 mm/sec (Diagnostic)	0.02 sec (20 ms)	0.10 sec (100 ms)	Reveals atrial flutter waves buried in 2:1 conduction block at 25 mm/sec.

9. Automated Classification and Performance Validation

Computer-generated rhythm classification statements are adjuncts to clinical practice, not substitutes for it. Physician overreading is a non-negotiable requirement for validation of all AI-generated ECG statements. The automated classification system provides decision support that must be continuously underwritten by the trust measurement framework and subject to deterministic veto by the circuit breaker at every output cycle.

9.1 Arrhythmia Classification Framework: AAMI Mapping

The system maps the 14 MIT-BIH rhythm classes into the five AAMI general categories to standardize classification terminology across clinical platforms and training datasets. This taxonomy also defines the circuit breaker’s ontological boundaries for plausible classification assertions.

Code	Category	Included Beat Types	CB Constraint
N	Normal	Sinus Normal, LBBB, RBBB, Atrial Escape, Nodal Escape	QRS duration within physiological limits for subtype.
S	Supraventricular Ectopic	APC, Aberrated APC, Blocked APC, Nodal/Junctional APC	P-wave morphology consistent with supraventricular origin.
V	Ventricular Ectopic	PVC, Ventricular Escape, Ventricular Flutter	QRS > 120 ms; no preceding P-wave with fixed relationship.
F	Fusion	Fusion of ventricular and normal beats	Intermediate morphology verified against N and V templates.
Q	Unclassified	Beats not meeting criteria for N, S, V, or F	Automatic physician overread required; CB flags low-confidence.

9.2 Advanced Signal Analysis Techniques

- Wigner-Ville Distribution (WVD):** The system must convert the ECG into an Analytical Signal via the Hilbert Transform before applying WVD. This preprocessing step is mandatory to eliminate negative-frequency cross-terms that would otherwise create pseudo-energy artifacts in the time-frequency plane, which the AI layer would interpret as transient high-frequency events.
- Higher Order Statistics (HOS):** The system computes Skewness (3rd-order cumulant, measuring signal asymmetry) and Kurtosis (4th-order cumulant, measuring peakedness

relative to a Gaussian) for each beat segment. These higher-order statistics provide a nonlinear, noise-robust representation that can detect non-Gaussian artifacts arising from electrode motion or electrosurgical interference.

10. AI Circuit Breaker: Multi-Layer Validation Logic

The AI Circuit Breaker is the central enforcement mechanism of the entire trustworthiness framework. It operates as a cross-cutting architectural component spanning all four processing layers and all six subsystems. Its authority is absolute: it holds the deterministic veto over every AI-generated assertion about cardiac rhythm, regardless of the AI model’s expressed confidence level. A Super-Huberous assertion—one with high confidence but lacking physical grounding—is treated as the highest-risk failure mode.

10.1 Component Architecture

Component	Function	Source of Truth
LLM (AI Classification Engine)	Strategic: Generates high-level rhythm classification statements, ST interpretations, and clinical recommendations.	Probabilistic inference over learned feature distributions from validated training dataset.
SLM (Small Language Model)	Tactical: Evaluates local context including beat-by-beat variation, sensor state, and real-time template consistency.	Short-range data patterns from the real-time ECG buffer and environmental sensor feeds.
Deterministic Sensors (ECG / ADC)	Ground Truth: Provides immediate, physics-based feedback on the actual cardiac electrical signal from the patient.	Physical electrochemical signal governed by immutable laws of cardiac electrophysiology.
Circuit Breaker Logic	Central Control: Evaluates the Intent Delta between AI output and sensor ground truth; enforces deterministic veto.	Deterministic veto rules encoded in W3C RDF/OWL cardiac domain ontology.

10.2 The Core Trip Logic

The operational logic of the circuit breaker is defined by the following deterministic evaluation, executed on every AI classification output before it is transmitted to the clinical interface or fed back into the training pipeline. The evaluation is atomic and synchronous—the AI output is withheld until the circuit breaker completes its assessment.

```
CIRCUIT BREAKER TRIP LOGIC - ECG-Specific Implementation: IF (Intent_Delta >
Sensor_Ground_Truth_Tolerance) OR (Trust_Metric < Minimum_Trust_Threshold)
OR (Semantic_Anomaly_Score > Anomaly_Ceiling) OR
(Contextual_Relevancy_Index < CRI_Minimum) THEN ACTIVATE CIRCUIT BREAKER
(DETERMINISTIC VETO) EXECUTE Safe_State_Action(severity_level) FLAG
data_epoch for SOP-03 exclusion from training queue LOG: {timestamp,
AI_assertion, physical_signal_epoch, intent_delta, trust_scores,
filter_state, action_taken} ELSE TRANSMIT AI_assertion to clinical interface
ATTACH trust_score_metadata to output record
```

10.3 ECG-Specific Threshold Parameters

Parameter	Threshold	Clinical Basis
Ground Truth Tolerance (Amplitude)	±10 µV	AHA fidelity mandate for amplitude accuracy across all clinical scenarios.
Ground Truth Tolerance (Interval)	±10 ms	AHA recommendation for temporal coherence in simultaneous 12-lead systems.
Minimum Trust Metric Threshold	0.65 (0–1.0 scale)	Below this level, AI confidence is insufficient for autonomous clinical action.
Semantic Anomaly Score Ceiling	0.40 (0–1.0 scale)	Above this level, divergence from cardiac ontology indicates hallucination.
CRI Minimum	0.60 (0–1.0 scale)	Below this level, environmental conditions are insufficient to ground the AI assertion.
SOP-02 Lockout Trigger	3 trips / 10 min	Pattern of repeated failures requires human trustee intervention before restart.

10.4 Three-Level Safe State Activation

- 4. **Level 1 — Soft Alert:** AI classification is flagged with a low-confidence warning tag and transmitted with a mandatory physician overread indicator. Clinical action may proceed, but must be accompanied by cardiologist review.
- 5. **Level 2 — AI Hold:** AI classification is withheld from the clinical interface. Raw ECG signal and computed measurements (intervals, amplitudes) are transmitted without AI interpretation. A cardiologist's review is mandatory before any clinical action.
- 6. **Level 3 — Emergency Halt:** All autonomous AI operations cease. The device enters a safe monitoring state, continuing to record and display the raw ECG signal. Manual human control is required to resume.

SOP-02 — Triple-Trip Lockout Protocol

If the Deterministic Veto is activated more than 3 times within any 10-minute window, all autonomous AI operations are locked out until a designated human Trustee reviews the full CB event log, identifies the root cause, and provides authenticated electronic authorization to resume. The Human-AI Calibration Coefficient is updated before autonomous operation restarts.

11. Semantic Governance: The Knowledge Fabric Architecture

Semantic governance is the architectural mechanism that industrializes clinical knowledge into machine-readable, verifiable rules. For an ECG system, this means encoding the complete domain of clinical cardiology knowledge—including the boundaries of physiological possibility—into a formal ontological structure that the circuit breaker can query in real time. This Knowledge Fabric is the Immutable Truth layer of the system.

11.1 Design-Time Semantic Guardrails: Encoding Impossible States

W3C RDF/OWL ontologies formalize the semantic constraints defining the boundaries of physiological possibility. These are hard-coded logical rules representing established clinical knowledge—not probabilistic models. No AI inference may contradict these constraints without triggering an immediate circuit breaker event.

Constraint Category	Example ECG Impossible State Definitions
Heart Rate Bounds	Intrinsic ventricular rate > 300 bpm without documented pre-excitation is physiologically impossible. Any AI classification asserting this triggers immediate CB activation at Level 3.
Conduction Interval Limits	PR interval < 60 ms without documented pre-excitation is an impossible state. QRS duration < 60 ms in adults is an impossible state. QT > 700 ms triggers a mandatory medication-context check.
ST-Segment Logic	Simultaneous ST elevation in all 12 leads without a pericarditis context is classified as a sensor artifact state pending CRI evaluation, not a primary pathological classification.
Rhythm Logic Constraints	P-wave before every QRS with constant PR is logically exclusive to second-degree and complete AV block. Concurrent assertion of both is an ontological violation triggering CB activation.

11.2 Knowledge Fabric Technology Stack

Component	Technology Platform	ECG System Role
Ontology Management	Metaphactory with OMG ODM (Ontology Definition Metamodel)	Models clinical cardiology concepts; manages version control for ECG domain ontology updates.
Knowledge Graph Storage	GraphDB — High-performance RDF triple store	Stores all cardiac ontology rules; performs real-time inference against incoming AI assertions.
Intent Pattern Storage	Domain-Specific Vector Databases	Captures ECG feature patterns; supports semantic similarity search for classification context validation.

Analytics Workbench	Tom Sawyer Perspectives (centrality, clustering, pathfinding)	Root cause analysis of CB event clusters; impact visualization for misclassification propagation.
Integration Protocol	Model Context Protocol (MCP)	Enables identical Trust Engine deployment across manufacturing R&D, IoT edge devices, and cloud clinical dashboard.

11.3 Inbound Data Validation Workflow

7. **Request:** A new data asset, annotation, or knowledge extension is queued for integration with full provenance metadata.
8. **Retrieval:** Extraction of the digital asset, its URI, schema metadata, acquisition parameters, filter state, and electrode placement metadata.
9. **RDF Generation:** Serializes the ECG record and metadata into W3C RDF triples aligned with the cardiac domain ontology, using standard URIs.
10. **Validation:** Structural checks against SHACL constraints. Any record associated with a prior CB trip is automatically rejected (SOP-03 enforcement).
11. **Loading:** Validated assets loaded into the Graph Store with full versioning and KGCL (Knowledge Graph Change Language) audit trail.

12. Trust Metrology Dashboard: Measurement Models and Metrics

The Trust Metrology Dashboard is the central mission control interface for the trustworthiness engineering system. It synthesizes real-time AI accuracy metrics, data integrity rejection rates, sensor health indicators, and human operator status into a unified operational view—providing Engineered Trust: the shift from blind reliance on AI to informed, quantifiable confidence in autonomous cardiac analysis.

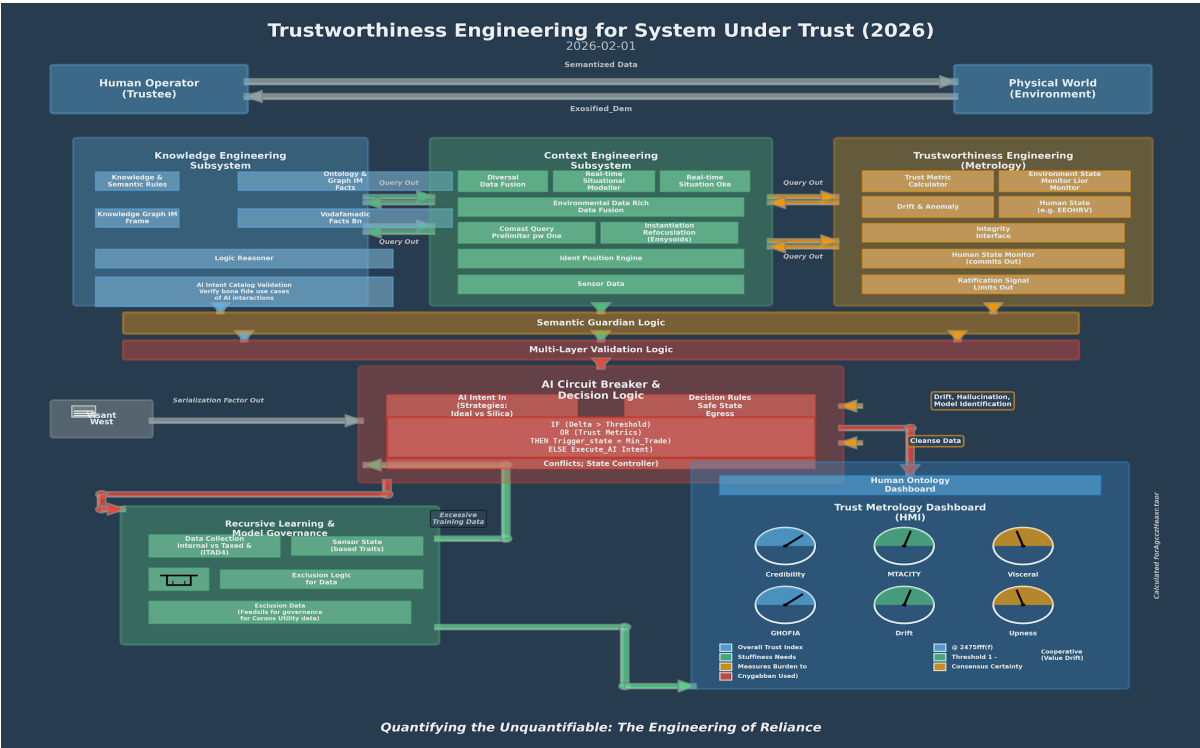


Figure 12.1 — Trust Metrology Dashboard: Real-Time Gauges for Credibility, Validity, Viability, Overall Trust Index, Scale, and Human Operator Status

12.1 Primary Trust Metrics

Metric	Definition	ECG Clinical Application
Mean Time Between Hallucinations (MTBH)	Average operating time before a high-confidence error that would have triggered a CB event without the veto.	Benchmarks AI reliability in arrhythmia detection. Declining MTBH signals model drift and triggers mandatory re-validation.
Value-Drift Coefficient (VDC)	Numerical measure of distance between current agent action and the original clinically validated intent over time.	Detects systematic shifts in classification behavior (e.g., increasing false-positive AFib rate) before patient safety events occur.

Semantic Consistency Score (SCS)	Measures alignment of new AI classification outputs with the approved core semantic assets in the cardiac ontology.	Tracks whether updated model versions produce classifications consistent with established clinical cardiology ontological constraints.
AI Suggestion Adoption Rate (ASAR)	Percentage of AI classification statements accepted by physician reviewers without manual correction or override.	Measures practical clinician trust in AI output. Declining ASAR indicates emerging discordance between AI behavior and clinical consensus.
Semantic Anomaly Score (SAS)	Real-time quantification of logical divergence between the AI interpretation and the established ECG domain ontology.	Primary CB evaluation metric. Elevated when AI asserts physiologically impossible states or contradicts ontological constraints.
Contextual Relevancy Index (CRI)	Evaluates whether the AI model is properly grounded in real-time environmental and sensor data.	Reduced by electrode impedance failures, powerline interference, motion artifact, or active monitoring filter during diagnostic recording.

12.2 Dashboard Gauge Indicators

The Trust Metrology Dashboard presents six real-time gauge indicators corresponding to the core dimensions of system trustworthiness. Each gauge operates on a 0–1.0 normalized scale with color-coded alert thresholds.

Gauge	Measurement Domain	Green / Alert / Trip	CB Action
Credibility	AI classification accuracy relative to physician overread consensus.	> 0.85 / 0.65–0.85 / < 0.65	L1 Alert / L2 Hold
Validity	Ontological consistency of AI output against W3C RDF/OWL cardiac constraints.	> 0.90 / 0.70–0.90 / < 0.70	L2 Hold / L3 Halt
Viability	Operational health of the sensor subsystem: electrode contact, ADC integrity, filter state.	> 0.80 / 0.60–0.80 / < 0.60	L1 Alert / L2 Hold
Overall Trust	Composite: Credibility (0.35) + Validity (0.40) + Viability (0.25).	> 0.75 / 0.55–0.75 / < 0.55	L2 Hold / L3 Halt
Scale (Volume)	Analysis throughput and latency; high volume may stress the CB validation pipeline.	Within SLA / Degraded / Overload	Queue escalation
Operator (Human)	Human-AI Calibration Coefficient: estimates cardiologist cognitive load via physiological proxy.	Optimal / Elevated / Impaired	Threshold adjust

13. Standard Operating Procedures

The three Standard Operating Procedures below constitute the governance layer of the AI Circuit Breaker. They define the mandatory initialization sequence, the continuous calibration protocol, and the data exclusion policy, which collectively maintain the integrity of the AI pipeline throughout the ECG device's complete operational lifecycle.

13.1 SOP-01: Initialization and Semantic Handshake

Before the ECG device begins any autonomous operation, the system must execute the Semantic Handshake protocol. This is the mandatory pre-flight check that verifies all preconditions for trustworthy operation are met. If any step fails, autonomous AI operation is prohibited.

12. **Step 1 — Knowledge Graph Query:** Ping the Knowledge Graph (GraphDB) to verify the local ECG ontology is current, consistent, and responsive.
13. **Step 2 — Rule of Ten Precision Check:** Verify that ADC resolution is at least one order of magnitude greater than the operational tolerance threshold ($\leq 1 \mu\text{V}$ for a $10 \mu\text{V}$ AHA mandate).
14. **Step 3 — Electrode Impedance Validation:** Automated impedance sweep across all 10 electrodes. All impedances must be $< 5 \text{ k}\Omega$. Lead-off flags must be clear.
15. **Step 4 — Filter State Verification:** Confirm diagnostic filters are active: LPF 150 Hz (adult) / 250 Hz (pediatric); HPF 0.05 or 0.67 Hz (zero-phase verified). Monitoring filters must not be active.
16. **Step 5 — Trust Baseline Initialization:** Initialize all trust gauges to calibrated baseline values. Log full initialization state with timestamp and verified parameter set.

13.2 SOP-02: Real-Time Trust Calibration

- **Continuous Monitoring:** The Value-Drift Coefficient is computed on a rolling 10-minute window. Sustained drift above 0.15 units from the SOP-01 baseline triggers a Level 1 Alert.
- **Triple-Trip Lockout Rule:** If the Deterministic Veto is activated more than 3 times within any 10-minute window, all autonomous AI operations are locked out immediately.
- **Trustee Re-Calibration:** A designated human Trustee must review the full CB event log, identify the root cause, and provide authenticated electronic authorization before autonomous operation resumes.

13.3 SOP-03: Recursive Loop Sanitization (Data Exclusion)

SOP-03 is the immunological defense against Model Collapse—the recursive reinforcement of errors through the AI training pipeline. A model allowed to train on its own hallucinations progressively diverges from clinical reality while simultaneously increasing its expressed confidence.

- **Sanitization Filter (Continuous):** All ECG telemetry is tagged with the originating session's CB status. Any epoch associated with a CB activation is automatically flagged at the point of generation.
- **Trip-Flagged Data Purge:** Any data epoch associated with a CB activation is excluded from the training queue, including the triggering segment plus a preceding 30-second buffer, and all AI outputs generated during the flagged window.
- **Ontological Gating:** Before any update to the domain-specific vector database, the logic reasoner evaluates consistency with W3C RDF/OWL ontological rules. Updates that would cover ontologically impossible states are rejected.

14. Lifecycle Management and Reusability

One of the defining architectural advantages of this framework is its reusability across the complete digital lifecycle of the ECG device and its AI components. By separating the architecture into four distinct layers and managing all inter-layer communication through a standardized Model Context Protocol (MCP), the **Identical Trust Engine** can be deployed across fundamentally different operational contexts without re-engineering the core governance logic.

Deployment Context	Circuit Breaker Function	Trust Metrology Focus
Design Time (R&D)	Validates training data quality; enforces ontological constraints on annotations; prevents ingestion of impossible synthetic data.	MTBH baseline establishment; SCS calibration; reference VDC profile generation.
Run Time — Edge (IoT)	Real-time veto on AI rhythm alerts; SOP-01 handshake at power-on; CRI-driven suppression during motion artifact events.	Lightweight CRI/SAS computation; operator physiological monitoring via wearable HRV proxy.
Run Time — Clinical	Supervisory validation of batch AI interpretations; retrospective trip log analysis; physician overread workflow integration.	Full six-gauge Trustworthiness Dashboard; ASAR tracking; Trustee escalation workflow.
Continuous Learning	SOP-03 sanitization enforcement; ontological gating of all dataset updates; SHACL validation before graph modification.	VDC monitoring across model versions; SCS tracking for ontological drift in updated classifiers.

14.1 Human-AI Calibration Coefficient

A distinctive feature of this framework is its explicit recognition that the human operator is not a passive observer but an active, measurable component of the trust system. The Human-AI Calibration Coefficient monitors the clinical operator's cognitive state. It dynamically adjusts circuit breaker thresholds to ensure that the demand on human oversight never exceeds the operator's reliable review capacity.

- Physiological State Monitoring:** In high-acuity settings, HRV and EEG proxy metrics directly estimate cognitive load. In standard reading environments, the coefficient is estimated from operational proxies: studies pending review, time since last break, alert frequency, and trends in override rate.
- Threshold Adjustment Logic:** When estimated cognitive load exceeds a defined threshold, CB alert thresholds become more conservative, routing additional studies to Level 1 Alert status rather than autonomous clearance.
- Human as Ultimate Dimension of Trust:** The Human-AI Calibration Coefficient formalizes the principle that the human operator always holds absolute authority. No level of AI confidence can remove the human from the critical path of clinical decision-making in cardiac diagnostics.

15. Code of Ethics for Autonomous ECG Agents

The governance of autonomous agents in safety-critical clinical environments requires an explicit ethical framework defining the non-negotiable boundaries of machine autonomy. The three principles below are implemented as architectural constraints—verified at system initialization and monitored continuously during operation.

Ethical Principle	Technical Implementation	CB Enforcement
Observability Over Autonomy	No agent shall take a clinical action it cannot explain through its internal decision-trace logs. Every AI classification is accompanied by a full provenance record: input features, model version, trust scores, and ontological consistency check result.	Any classification without a complete provenance record is treated as an unverifiable assertion and blocked from clinical output.
Honesty in Uncertainty	Agents must signal uncertainty by providing confidence intervals and trust metric scores with every output. Unbounded high-confidence assertions are treated as architectural violations, not virtuous performance.	SAS and CRI thresholds serve as the quantitative implementation. The gap between AI confidence and trust metric score detects a super-huberous assertion.
The Human Veto	Human oversight is the final and non-negotiable dimension of trust. The cardiologist holds absolute authority to override any AI classification. The system is architected to make human override trivially easy and AI override of human decisions technically impossible.	Level 3 Emergency Halt can only be lifted by a human Trustee. No automated recovery protocol can bypass the human authorization step.

16. Engineering Implementation Checklist

The following checklist consolidates all mandatory and recommended implementation requirements across the complete architecture. All [REQUIRED] items must be verified and formally signed off before the system is authorized for clinical deployment.

16.1 Signal Acquisition and Processing

- ☐ [REQUIRED] Front-end oversampling at 1,000–15,000+ Hz achieved and verified.
- ☐ [REQUIRED] Processing minimum 500 Hz sampling for 1% amplitude fidelity confirmed.
- ☐ [REQUIRED] ADC resolution $\leq 1 \mu\text{V}$ (Rule of Ten for 10 μV AHA tolerance) verified at initialization.
- ☐ [REQUIRED] Pacemaker pulse detection: front-end min 10,000 samples/sec; no artificial amplitude spikes.
- ☐ [REQUIRED] Pacemaker markers: dedicated display row; markers distinguish pacing from intrinsic QRS.

16.2 Digital Filtering

- ☐ [REQUIRED] High-pass: 0.05 Hz standard; 0.67 Hz permissible with verified zero-phase bidirectional implementation only.
- ☐ [REQUIRED] Low-pass: 150 Hz (Adult/Adolescent); 250 Hz (Infant/Child).
- ☐ [REQUIRED] 40 Hz filter: mandatory user alert when active during diagnostic recording; auto-restore to diagnostic cutoff.
- ☐ [REQUIRED] Filter state (cutoff values, zero-phase flag) logged in all CB event records.

16.3 Waveform Synthesis and Measurement

- ☐ [REQUIRED] Time coherence: temporal alignment ± 10 ms accuracy across all 12 leads.
- ☐ [REQUIRED] Template construction: cross-correlation beat alignment; dominant/nondominant beat differentiation.
- ☐ [REQUIRED] Global interval measurement: earliest onset / latest offset logic for all waveform durations.
- ☐ [REQUIRED] Normal range recalibration: AV block and QRS interval ranges redefined for global measurement.

16.4 Data Compression and Storage

- ☐ [REQUIRED] AHA fidelity mandate: maximum compression error $< 10 \mu\text{V}$ from original signal on retrieval.
- ☐ [REQUIRED] Integrity check: post-decompression CB verification before any AI classification of archived records.
- ☐ [RECOMMENDED] Lossless compression: mandated future state where bandwidth and storage permit.

16.5 Clinical Hardware and Display

- [REQUIRED] Electrode placement: V1–V2 in 4th ICS; V4–V6 along strict horizontal plane through V4.
- [REQUIRED] Impedance sweep: automated at initialization; all electrodes < 5 kΩ; lead-off flags clear.
- [REQUIRED] Mason-Likar placement: torso ECGs metadata-tagged; AI loads modified reference ranges before classification.
- [REQUIRED] Display speed: 50 mm/sec support implemented for flutter and rapid arrhythmia identification.

16.6 AI Classification and Analysis

- [REQUIRED] AAMI mapping: MIT-BIH 14-class taxonomy mapped to N/S/V/F/Q with CB ontological constraints.
- [REQUIRED] Hilbert Transform: implemented prior to Wigner-Ville Distribution for all time-frequency analysis.
- [REQUIRED] Higher Order Statistics: Skewness (3rd order) and Kurtosis (4th order) computed for all beat segments.
- [REQUIRED] Physician overread: mandatory for all AI-generated clinical statements; non-overrideable by configuration.

16.7 AI Circuit Breaker and Trust Metrology

- [REQUIRED] SOP-01: Semantic Handshake executed (all 5 steps) at every device initialization.
- [REQUIRED] Trip logic: IF (Delta > Tolerance) OR (Trust < Min) OR (SAS > Ceiling) OR (CRI < Min) THEN ACTIVATE CB.
- [REQUIRED] Three-level safe state: Level 1 Alert / Level 2 AI Hold / Level 3 Emergency Halt implemented and tested.
- [REQUIRED] SOP-02: Triple-trip lockout within 10-minute window; Trustee re-authorization required before autonomous restart.
- [REQUIRED] SOP-03: All trip-flagged data epochs excluded from training queue; 30-second preceding buffer purged.
- [REQUIRED] Cardiac ontology: W3C RDF/OWL impossible states encoded; SHACL validation active on all data ingestion.
- [REQUIRED] Six-gauge dashboard: Credibility, Validity, Viability, Overall Trust, Scale, Operator all operational.
- [REQUIRED] MTBH / VDC / SCS / ASAR: all four primary metrics computed continuously and logged.
- [REQUIRED] Event logging: all CB events logged with full provenance (timestamp, assertion, epoch, delta, scores, action).
- [REQUIRED] Human-AI Calibration Coefficient: cognitive load estimation active; threshold adjustment logic enabled.

17. Conclusion: Quantifying the Unquantifiable

The architecture defined in this Technical Design Document represents a fundamental shift in the engineering of AI-assisted cardiac diagnostics. It is not sufficient to build an AI system that classifies arrhythmias with high accuracy under controlled conditions. It is necessary to build a system that maintains trustworthy behavior under all operational conditions—including degraded sensor states, novel arrhythmia patterns, model drift over time, and fatigued human oversight.

The AI Circuit Breaker does not constrain the capability of the AI system. It grounds it. By maintaining a continuous, quantified measure of the gap between what the AI asserts and what the physical cardiac signal shows, the circuit breaker enables the AI to operate at the boundary of its reliable performance envelope while ensuring that excursions beyond that boundary are detected, halted, and corrected before they reach the patient.

The design draws together two previously separate engineering disciplines: the metrological precision standards of diagnostic electrocardiography and the trustworthiness engineering framework of the AI Circuit Breaker and Trust Engine. Their integration is not additive—it is multiplicative. The ECG's physical ground truth becomes the anchor for the Circuit Breaker's deterministic veto. The Trust Metrology Dashboard's six gauges serve as the clinical interface through which Engineered Trust is made visible, measurable, and actionable to the human cardiologist, who remains the ultimate authority in the system.

In cardiac diagnostics, where a single misclassification can mean the difference between life and death, the Engineering of Reliability is not an aspiration. It is the minimum standard of responsible design.